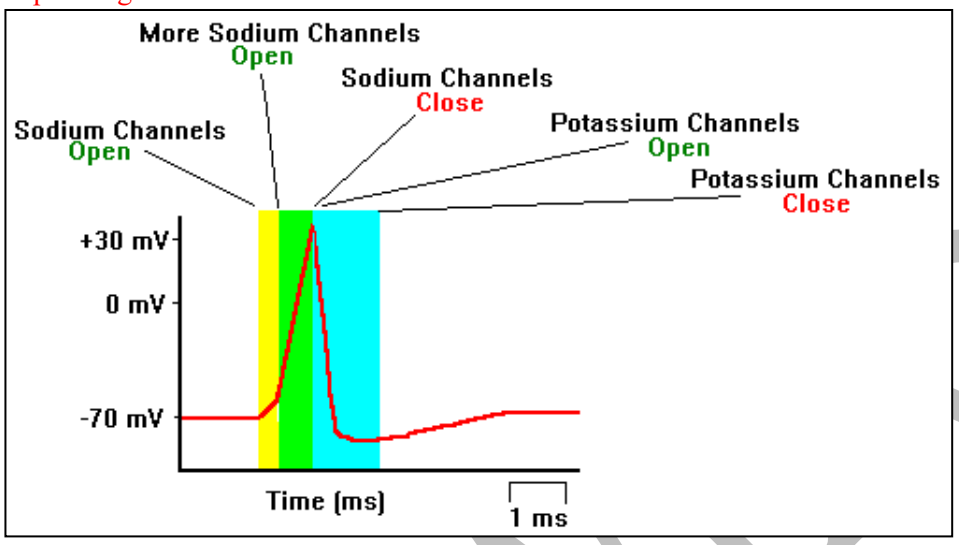
1. This question relates to the design of a multichannel instrument for *measuring bio-potentials from the muscles.*

a) Explain how a neural action potential is generated referring to the ionic flow of sodium and potassium ions. You are required to sketch the waveform of an action potential and annotate each phase, explaining the mechanism contributing to these.

[4]

[bookwork]

Explain figure shown



[Total 4 points]

b) Table 1.1 lists five commonly observed bio-potentials and their amplitude and frequency characteristics. A doctor requires an instrumentation system to record from a set of electrodes connected to the back muscles shown in Figure 1.1.

Design a complete instrumentation system for recoding bio-potentials from the back according to the following specifications:

Instrumentation Amplifier Gain: 21
Low-Pass Filter Roll off: 40dB/decade
High-Pass Filter Roll off: 60dB/decade
Supply Range: -5V to 5V

State all assumptions and values of resistors and capacitors used. You may assume that the instrumentation system utilizes the whole dynamic range available and 10nF capacitors are available for the filters.

Using table below for EMG:
High pass cut off: f_{HP}=20Hz
$R_{HP}=1/2\pi Cf_{HP}$ = 1/2π(10nF)(20Hz) =0.795 MΩ [1 point]
Low pass cut off: f_{LP}=2KHz
$R_{LP}=1/2\pi Cf_{LP}$ = 1/2π(10nF)(2KHz)=7.95KΩ [1 point]

Low pass 40dB/decade: Sallen Key can be used
High pass 60dB/decade: Opamp with RC + Sallen Key [1 point]

Instrumentation amplifier gain:
$V_{OUT} = -(V_1 - V_2)(1 + 2R_2/R_1)(R_4/R_3)$
For gain of 21 we select R4=R3=R2= 100 KΩ and R1= 10 KΩ [1 point]

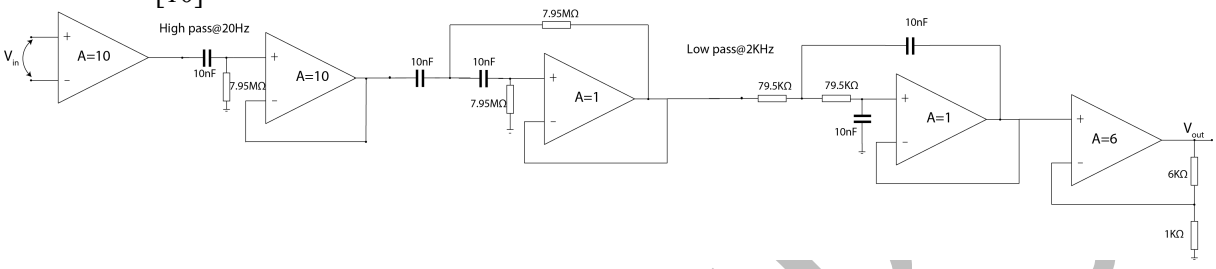
Supply Range: -5V to 5V

ANSWERS

VDD=5V, GND=0V, VSS=-5V
Max EMG = 100mV (from table)
V_{inptp} x A1 x A2=$10V_{ptp}$
A1=21
100mVx21xA2=10V
A2= 4.76 (gain of filter and final stage) [1 point]
Realised with a non-inverting amplifier with R1=9.5KΩ and R2=2KΩ [1 point]

A potential solution [4 points]:

[10]



c) The function of the system is to measure the muscle fibre conduction velocity (MFCV), determined by the propagation delay between the signals measured between electrodes 1-2 and 3-4. Electrode 5 is used to set the common mode voltage of the body. Design a system using your front end designed in 1.b which is able to measure the MFCV and output a digital number representing this value. State all assumptions in your design.

[6]

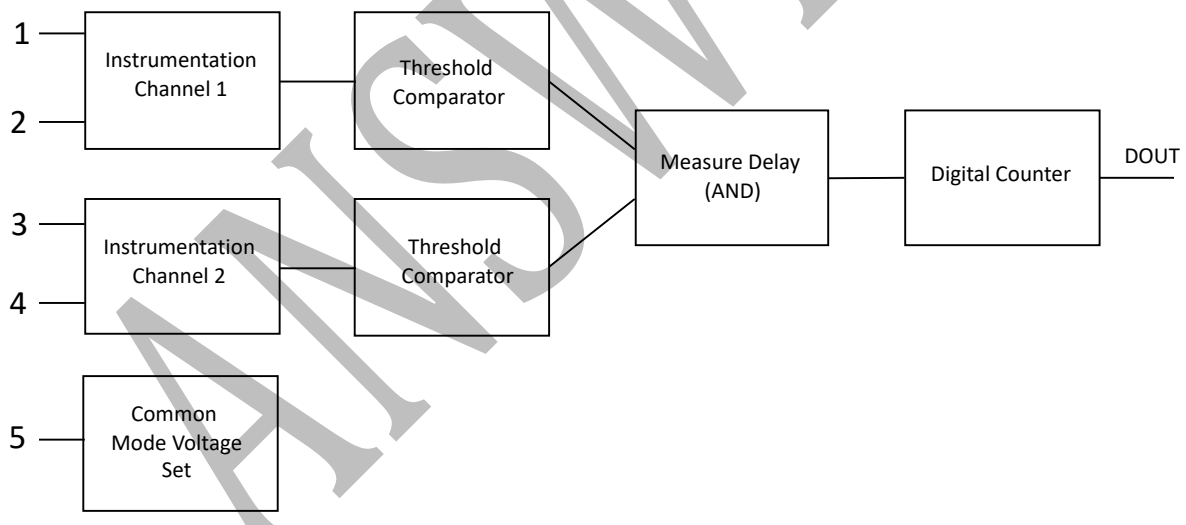


Table 1.1

Signal	**Frequency range (Hz)**	**Amplitude range (mV)**
ECG	0.1 – 300	0.05 – 3
EEG	0.5 – 40	0.001 – 1
EMG	20 – 2000	0.001 – 100
Neural Spikes	300-5000	0.001– 0.5
Local Field Potentials	10-200	0.001 – 5

ANSWERS

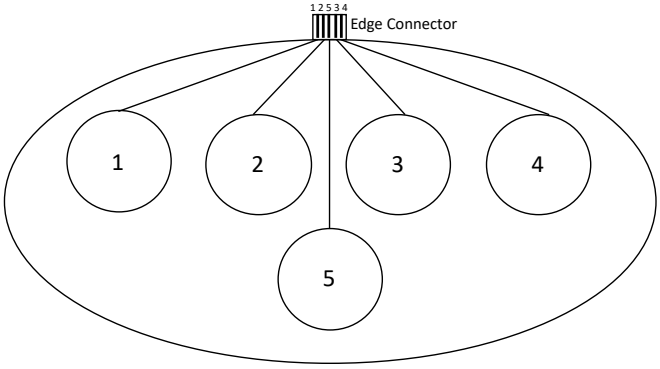


Figure 1.1

2. This question relates to electrical stimulation applied to bioelectronic medicines:

Bioelectronic medicines are implantable devices that modulate peripheral or autonomic nerves (such as the cervical vagus nerve) to alter end-organ function, yielding therapeutic benefits in systemic inflammatory disorders (e.g., rheumatoid arthritis, Crohn's disease).

(a) List the main electrode types used for peripheral-nerve stimulation. For each, state one advantage and one limitation. **[5]**

- Epineural cuff electrodes wrap around the nerve without piercing it, so they cause little chronic damage and are easy to explant; however, because current spreads circumferentially they give poor fascicular selectivity unless highly segmented. [2]
- Intrafascicular wires such as TIME or LIFE leads go inside a fascicle, giving much better fibre-group selectivity, but the act of penetrating the perineurium and the subsequent micromotion increase the risk of fibrosis and long-term signal loss. [2]
- Penetrating micro-electrode arrays (e.g., Utah arrays) can address individual axons and even record while stimulating, yet they demand the most invasive surgery and their tiny sites are severely charge-density-limited. [1] (this is less relevant to PNS).

(b) The human cervical vagus nerve is approximately 5 mm in diameter. Sketch a cuff-based electrode whose geometry increases stimulation selectivity relative to a simple two-ring cuff and explain briefly why this design is more selective. **[5]**

- To improve selectivity, replace the usual two continuous rings with one ring subdivided into eight independent electrode segments [1]
- that together match the nerve's inside circumference, assuming average of ($5\text{mm} \times \pi \approx 16\text{mm}$) [1].
- A practical pattern is 1.5mm metal pad + 0.5mm separation, repeated eight times ($8 \times 1.5\text{mm} + 8 \times 0.5\text{mm} = 16\text{mm}$). Each segment is 1mm wide axially, giving an individual contact area of about 1.5mm^2 while the 0.5mm gap provide electrical and mechanical isolation [1].
- During stimulation you drive one segment as the cathode and either of its two neighbours to the return node (anode) [1],
- so only fascicles nearest that sector reach threshold, whereas a conventional two-ring cuff lets current wrap around the whole nerve and excites many fascicles and fibres indiscriminately. [1]

(c) Using the smallest active contact area proposed in your electrode design in part (b), assess the viability of the electrode being used to deliver a biphasic, cathodic-first stimulus of 1mA with a 800µs phase width. Assume platinum-iridium has a reversible charge-density limit of $200\mu\text{Ccm}^{-2}$ and that the fabrication process yields an interface impedance of $12\Omega\text{cm}^2$ at 1kHz.

(i) Estimate the maximum cathodic current the contact can safely deliver. **[2]**

ANSWERS

- With a 1 mm² platinum–iridium contact (≈ 0.01 cm²) and a reversible charge-density ceiling of 200 µC cm⁻² [1], the most charge the cathodic phase may safely deliver is: Qmax=200uC/cm2 x 0.015cm2=3uC [1]

- Dividing by the specified 800 µs phase width gives: Imax=Qmax/t=3uC/0.8ms=3.75mA. [1]

(ii) Calculate the minimum voltage compliance the stimulator must provide. **[3]**

- With a specific interface impedance of 12Ωcm², a 1.5mm² contact (0.015cm²) presents an electrode impedance of 800Ω. Because current passes through both the cathode and the anode, the two contacts together contribute about 2x 800Ω = 1.6kΩ. [2]
- Adding an estimated 1kΩ of tissue and lead resistance gives a total load of roughly 2.6kΩ, so a 1mA stimulus would require a compliance voltage of approximately 2.6V (outside the circuit). [1]

(d) Based on the compliance calculated in part (c), outline the key requirements for the output stage of a current-mode neural stimulator. **[5]**

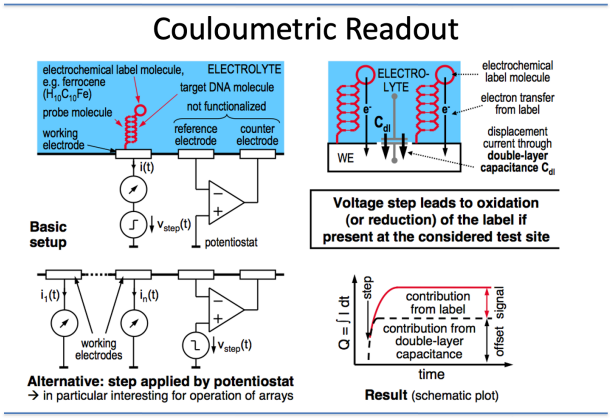
- Given the ≈ 2.6V worst-case load determined in part (c), the output stage must first run from a rail that still leaves at least this much across the electrodes after the output transistor stack drops its saturation (or cascode) voltage; i.e. a 4V supply would provide ~1.4V of "design headroom" for switch on-resistance, sensing circuitry, and any compliance-monitor comparator [2].
- Second, the stage has to look like an almost ideal current source: a small-signal output resistance ≥1MΩ is routinely achieved with a cascoded mirror; this keeps the programmed 1mA within a few percent while the node swings between 0V and 2.6V [2].
- Third, a series DC-blocking capacitor is essential to prevent any net Faradaic (DC) current [1].

ANSWERS

3. This question relates to the design of an instrument for detecting DNA.

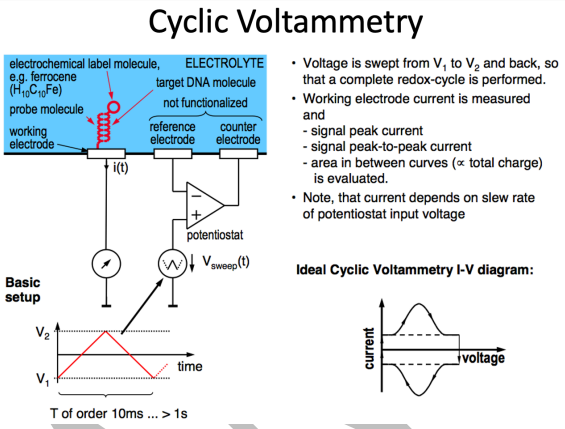
a) Explain two methods which can be used to detect DNA with an electrochemical sensor, describing the physical phenomena which allow these detection methods and stating one advantage and one disadvantage for each method.

Method 1:



Advantage: Can instrument to an array of sensors using one potentiostat.
Disadvantage: Requires labels on the DNA sample to generate a current.

Method 2:



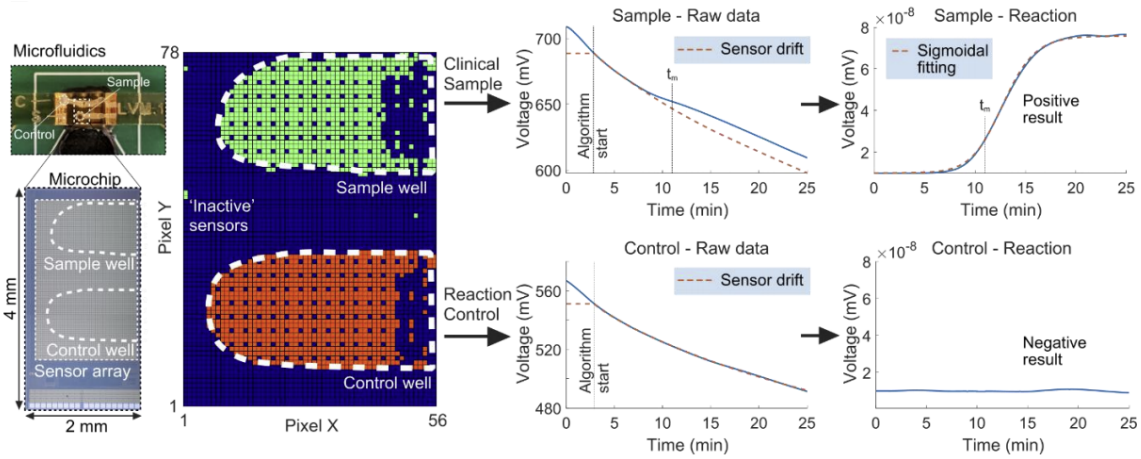
Advantage: Complete scan of the solution to see all redox species.
Disadvantage: Slower than amperometry as it requires triangular waveform with specific period T per measurement.

[6]

b) Figure 2.1 shows typical reaction measured from an ISFET sensor used for a bacterial test, where there is a change in pH caused by the DNA of the bacteria binding to its complementary target. The output voltage shown is linearly dependent on the pH change and the sensor has baseline drift. Figure 2.1a is a positive confirmation of the bacteria and Figure 2.1b is a negative control reaction to measure the baseline drift of the sensor. Given the data in Figure 2, design a suitable diagnostic system to measure bacterial DNA using ISFETs, sketching its system architecture and circuits for instrumentation.

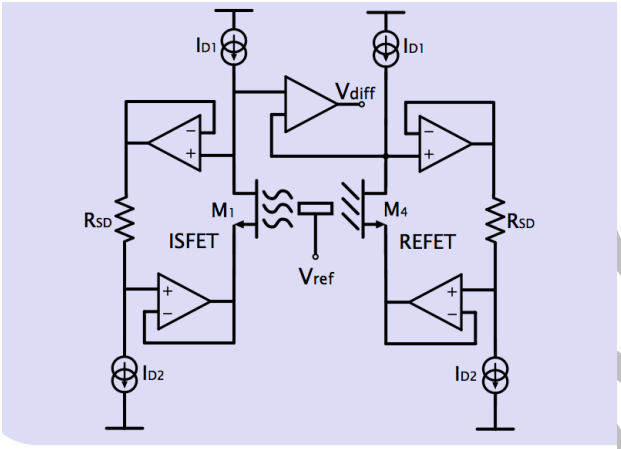
Student is expected to identify that you can run ISFETs in separate wells, one containing the probes to identify the bacterial DNA. Then design a differential instrumentation system to interface to these two wells.

ANSWERS



4 points.

Potential differential instrumentation:



Additional bonus given for baseline adaption.

4 points.

[8]

c) The circuit shown in Figure 2.2 is an advanced ISFET front end which outputs a signal V_{out} in the form of a pulse whose width is proportional to pH. Its operation is based on two phases, a reset phase when RST = 0 and a readout phase where RST = V_{DD}. The ISFET sensor M_0 is operating in strong inversion.

i. Derive the relationship between pH and the length of the output voltage pulse V_{out}, stating any assumptions made.

[Application of taught theory]

The circuit encodes the input in the time domain.

In the reset phase, M1 is on and V_a is charged up to V_{DD}.

In the readout phase, M1 is off so the C discharges through M0. M0 is biased in weak inversion so the current flowing the device is

$$I_{DS} = \beta(V_{GS} - V_T)^2$$

The discharge current for the capacitor C is

ANSWERS

$$I_{DS} = C\frac{dV_a}{dt}$$

Hence V_a discharges with a rate $I_{DS,0}/C$.

$$\frac{dV_a}{dt} = \frac{\beta(V_{GS} - V_T)^2}{C}$$

The inverters quantify the discharge as a pulse, encoding the input voltage in time. Assuming that the switching point of the inverters is at $\frac{V_{DD}}{2}$ (and that the discharge is linear), the pulse width is given by

$$\tau = \frac{V_{DD}}{2}\frac{C}{\beta(V_{GS} - V_T)^2}$$

$$V_T = V_{TH(MOS)} + V_{Chem}$$

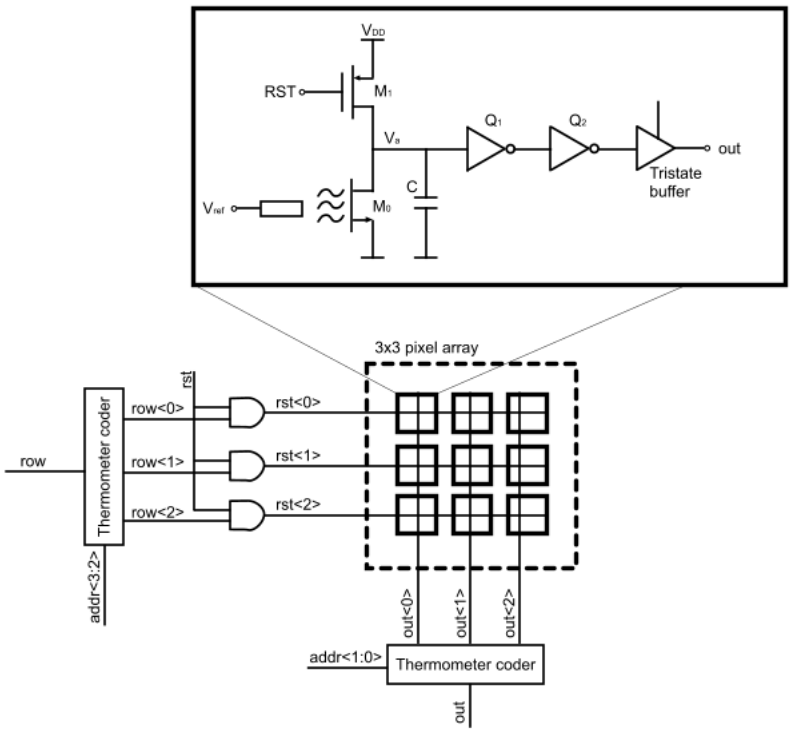
$$V_{Chem} = \gamma + 2.3aU_tpH$$

Grouping all constants, K, yields:

$$\tau = \frac{V_{DD}}{2}\frac{C}{\beta(K - 2.3aU_tpH)^2}$$

[3]

ii. Show how the architecture in Figure 2.2 can be used to instrument to a 3x3 array of ISFET sensors.



[3]

4. This question relates to a cochlear implant:

WERS

a) Explain why most commercial cochlear implants typically use a monopolar stimulation configuration rather than bipolar. **[3]**

- Shunting - The cochlea is fluid-filled, so the impedance between two adjacent intracochlear contacts is very low; in bipolar or multipolar modes most current shunts directly between the metal sites instead of crossing the spiral ganglion [1] ,
- Efficiency - so significantly more charge and a higher compliance voltage are needed for the same loudness. The extra drain shortens battery life and forces a bulkier power stage, whereas monopolar pairs each intracochlear contact with a remote return, raising the load impedance and maximising efficiency [1].
- Complexity - Monopolar drive also needs only one high-voltage current source per channel, fewer conductors, simpler packaging and placement, yet still yields clinically acceptable loudness growth, so manufacturers default to it. [1]

b) A behind the ear (BTE) sound processor feeds a 22-contact electrode array via an implanted receiver-stimulator.

(i) Sketch the basic architecture of the complete cochlear implant system. **[3]**

Sketch, depending on how well adapted textbook diagram is to specifics given above. Splitting function between external device and implant via inductive link [1], speech processor internal architecture [1], implant basic architecture [1].

(ii) List five key functional blocks between the microphones and the electrodes, and for each block state its principal design constraint (e.g. power, size, latency, EMC, safety, etc.) with a one-sentence justification. **[5]**

- Low-noise microphone pre-amp - must minimise power while keeping input-referred noise < 30 dB EIN. [1]
- Analysis filter bank - latency must stay below ≈ 2 ms to preserve audio-visual synchrony. [1]
- Automatic-gain-control/compression - safety-driven to keep neural current within patient's comfort window. [1]
- RF telemetry encoder/modulator - EMC limits radiated power, so it must meet < 100 µW EIRP while achieving low BER [1]
- High-voltage stimulation driver - safety and size dominate: hermetic packaging plus active charge-balancing logic must fit in the implant can [1]

c) A fully implantable research prototype uses a 3.7V, 15mAh Li-ion rechargeable micro-battery to power a custom chip that draws a steady 126µW. The battery is recharged periodically via the same 13.56MHz inductive link that carries calibration data. Calculate the maximum operating time, in hours, between charges if the 126µW demand is truly constant. State two reasons why the device is likely to consume more than stated, and one practical reason why the real-world operating time might be shorter than your calculated figure. **[5]**

- A 3.7V lithium-ion cell rated at 15mAh stores about 55.5 mWh of energy (simple product of voltage and capacity) [1].
- Dividing by the chip static 126µW consumption demand yields an ideal run-time of ≈ 440hr, which, if the implant is worn 15hr per day, translates to roughly 24 days between charges [1].
- That figure is optimistic: the stimulation current rises with ambient sound, so average power climbs in noisy settings [1].
- Real-world life is further shortened by housekeeping loads such as standby telemetry, sensor bias, self-discharge and inductive-charging losses [1].

ANSWERS

- Moreover, DC-DC converter inefficiency and the cell's cut-off voltage mean a fraction of the nominal 55.5 mWh is never delivered to the electronics, trimming endurance even more [1].

(d) Name two clinically used hearing-restoration devices other than the cochlear implant. For each device briefly describe its working principle, state one clear advantage it offers relative to a conventional cochlear implant and identify one engineering or clinical challenge that has limited its wider adoption. **[4]**

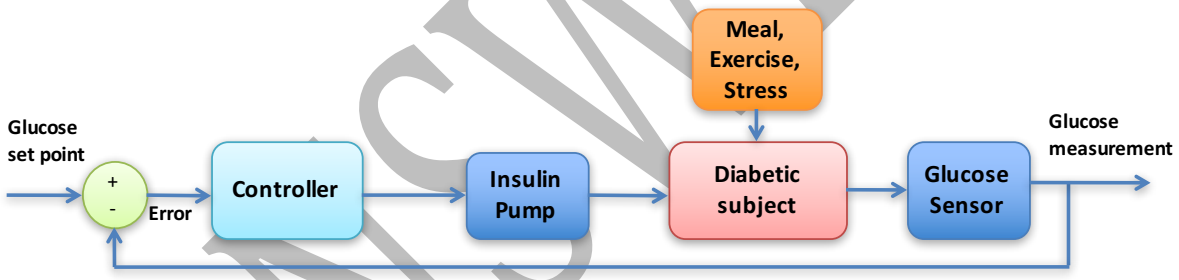
- Bone-anchored hearing aid (BAHA). Audio is converted to skull vibration via a titanium abutment, stimulating a normal cochlea in patients with conductive or mixed loss. Advantage: bypasses the damaged middle ear with minimal surgery. Challenge: percutaneous implant is prone to skin infection and offers limited high-frequency output. [2]
- Auditory brainstem implant (ABI). A paddle electrode rests on the cochlear nucleus to excite second-order neurons in people lacking an auditory nerve (e.g. NF-2 sufferers). Advantage: provides sound awareness when a cochlear implant is impossible. Challenge: delicate neurosurgery and poor place selectivity limit speech understanding. [2]

5. This question relates to the design of an artificial pancreas for treatment of diabetes.

a) Sketch a system level diagram of a closed loop system which works as an artificial pancreas and describe its operation highlighting three main disturbances to the system and two challenges.

Artificial Pancreas consists of continuous glucose sensor, insulin pump and controller.

Closed-loop diagram:



[5]

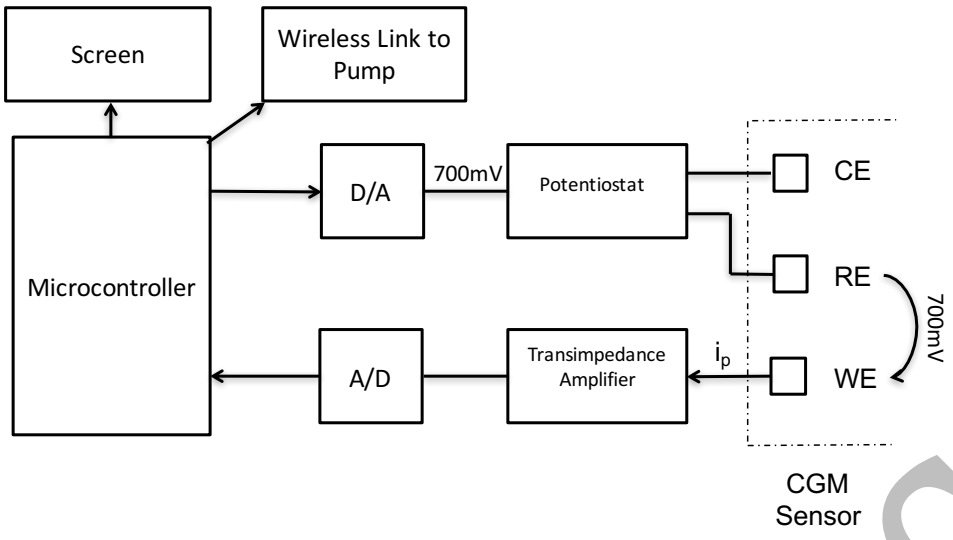
Explanation of operation (bookwork) highlighting Three disturbances: Meal, Exercise and Stress. [3]
One Challenge: CGM accuracy and delays due to the subcutaneous route of sensing and infusion. [2]

[10]

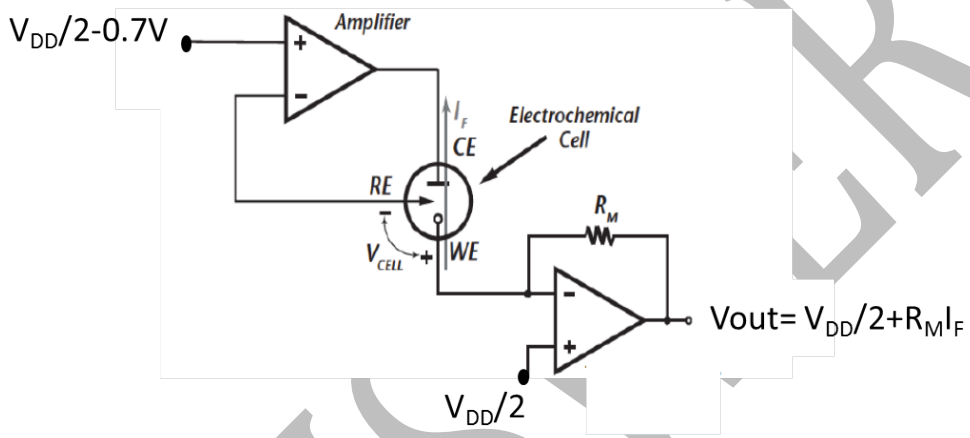
b) The blood glucose of a diabetic is expected to fluctuate in the range of 0-20mM. A continuous glucose sensor consists of an enzyme which can generate 10nA of current for every 1mM of glucose when biased with a redox potential of 700mV. Design a complete instrumentation system for an artificial pancreas to sense blood glucose using this enzyme which works off a 3V supply and maximizes the dynamic range available.

System:

ANSWERS



[5 points]
Instrumentation to the sensor:



[

V_{cell}= 0.7 V
VDD= 3V, $I_{F,max}$= 200nA
R_M*$I_{F,max}$=< 1.5V
Then R_M =< 7.5 MOhm
[5 points]

[10]

ANSWERS